Project Evaluation Results

Datamites Project Feedback Report

Team	PTID-AIE-JUL-26-11142	**Project**	PRAICP-1002
Course	AIE	**Overall Grade**	A
Topic	Traffic Sign Detection		

Dear Team Member,

Please find below the detailed evaluation feedback for your project submission. Review each section carefully and use the improvement suggestions to guide your next steps.

Data Understanding

Present Implementation

The submission provides a comprehensive analysis of the dataset, including class distribution, resolution variations, and detection of corrupt files. The use of memory-safe indexing to handle large datasets without loading them entirely into RAM is a significant strength. Visualizations and domain-specific observations regarding traffic sign characteristics (lighting, blur, perspective) are effectively connected to the modeling strategy.

Scope for Improvement

While the class imbalance analysis is thorough, adding a visualization specifically comparing the class distribution of the training, validation, and test sets after the stratified split would further confirm the integrity of the data partitioning.

Data Preprocessing

Present Implementation

The preprocessing pipeline is robust, covering resizing, normalization, and the handling of memory constraints. The choice to use a Rescaling layer within the model architecture is a best practice for production readiness, ensuring consistent normalization for future inference. The implementation correctly identifies that horizontal flipping is inappropriate for directional traffic signs, demonstrating high domain awareness.

Scope for Improvement

The submission could benefit from a brief discussion on why 64x64 was chosen as the target resolution and how it might impact the model's ability to distinguish fine details (like the numbers on speed limit signs), perhaps by comparing performance with a slightly higher resolution in future iterations.

Model Development

Present Implementation

The submission excels in model development by establishing a clear baseline CNN followed by a structured hyperparameter tuning process using Keras Tuner. The use of callbacks (EarlyStopping, ReduceLROnPlateau) and class weights to mitigate imbalance is technically sound. Performance is evaluated using a comprehensive suite of metrics including macro-averaged Precision, Recall, and F1-score, which are appropriate for this multi-class imbalanced problem.

Scope for Improvement

Although the submission compares a baseline to a tuned model, the architecture remains largely the same. The project would be strengthened by comparing the custom CNN against a lightweight pre-trained transfer learning architecture (e.g., MobileNetV2), which is standard practice in modern computer vision to achieve higher accuracy with less effort.

Results & Conclusion

Present Implementation

The results are presented clearly, with thorough analyses of the Confusion Matrix and a detailed gallery of misclassified images. The business impact is well-articulated, connecting the model's performance to real-world safety and autonomous vehicle applications. The discussion of failure cases in the confusion analysis provides valuable actionable insights for future improvements.

Scope for Improvement

The conclusion would benefit from a more concrete discussion on the limitations of the current classification-only approach compared to an end-to-end detection system (e.g., YOLO or SSD), particularly regarding how the current model would perform on a live, uncropped video stream.

Code Quality & Reproducibility

Present Implementation

The code is highly structured, readable, and includes clear commenting throughout. Variable naming is professional and consistent. The use of a fixed seed and the modularization of functions (like `evaluate_model` and `predict_sign`) ensure that the workflow is fully reproducible and professional in quality.

Scope for Improvement

No major improvements are required; the code is at an industry-standard level.

Overall Feedback

Present Implementation

This is an exceptional project submission. The student demonstrates a deep understanding of the entire machine learning lifecycle, from memory-efficient data handling to hyperparameter optimization and rigorous evaluation. The attention to domain-specific details—such as disabling horizontal flips due to the directional nature of traffic signs—indicates a level of critical thinking that exceeds the requirements of a typical capstone.

Scope for Improvement

The only meaningful avenue for further growth is exploring transfer learning architectures, as mentioned in the Model Development section, to see if state-of-the-art performance can be achieved on the more difficult-to-classify signage.

Best regards,

LMT